



NETWORK INTRUSION DETECTION SYSTEM

CODE ALPHA CYBER SECURITY INTERNSHIP

By Om Mehra



INTRODUCTION TO IDS



An Intrusion Detection System (IDS) monitors network traffic and detects suspicious activities or security threats.

The purpose of an IDS is to identify potential attacks and alert administrators before damage occurs.





PROJECT OBJECTIVES

- 01 Monitor network traffic in real time.
- 02 Detect suspicious activities.
- 03 Generate security alerts.
- 04 Improve network security awareness.
- 05 Demonstrate basic IDS functionality using Python.



TECHNOLOGIES USED

Programming Language:
Python

Library:
Scapy

Development Tool:
Visual Studio Code



Platform:
Windows



DETECTION RULES IMPLEMENTED

ICMP Flood Detection

Detects excessive ping requests.



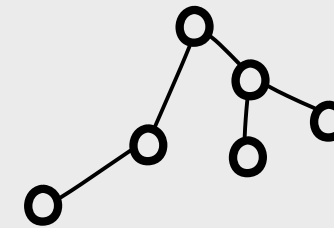
SYN Flood Detection

Detects excessive TCP SYN packets.

Suspicious Port Detection

Alerts on connections to risky ports such as:

- Port 21 (FTP)
- Port 23 (Telnet)



IDS WORKFLOW

Network Traffic



Packet Capture



Rule Analysis



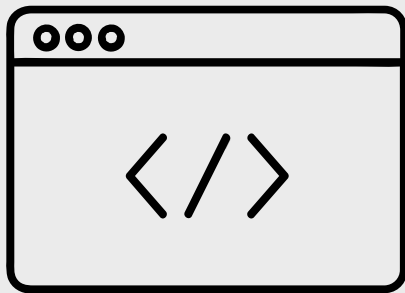
Threat Detection



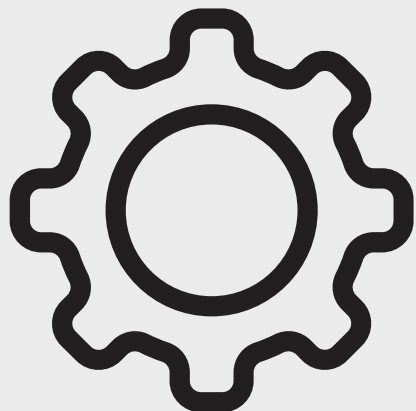
Alert Generation



IMPLEMENTATION



The IDS captures packets using Scapy and analyzes them in real time.



The system checks packets against predefined security rules and generates alerts when suspicious activity is detected.

DETECTION RESULTS

The IDS successfully generated alerts for:

- Excessive ICMP traffic
- Excessive SYN packets
- Suspicious port access attempts

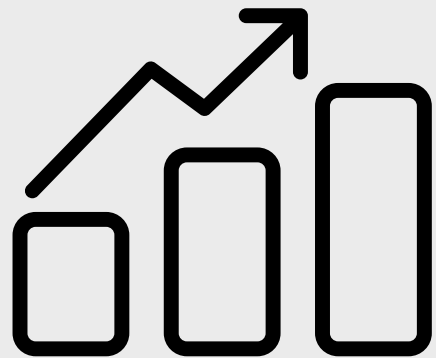
```
(venv) PS C:\Codealpha\codealpha_tasks\Task4_Network_Intrusion_Detection_System> python .\ids.py
Network Intrusion Detection System
Press CTRL + C to stop

[ALERT] Possible SYN Flood from 192.168.0.101
[ALERT] Possible SYN Flood from 192.168.0.101
```



ADVANTAGES OF IDS

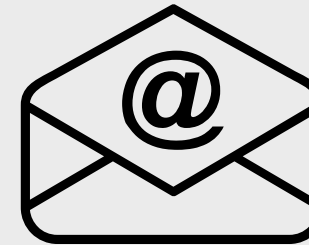
- Detects suspicious network activity.
- Provides real-time monitoring.
- Helps identify attacks early.
- Improves network visibility.
- Enhances security awareness.



FUTURE ENHANCEMENTS



- Email notifications
- Log file generation
- Graphical user interface
- Advanced attack detection
- Machine learning integration



CONCLUSION



This project demonstrates a basic Network Intrusion Detection System using Python and Scapy.

The IDS successfully monitors network traffic and generates alerts for suspicious activities.





THANK YOU

By Om Mehra